

Dynamic Website Development

CIS-241 30 O

Summer Term 2024-2025 School Year Section 30 O 3.00 Credits 05/19/2025 to 07/25/2025

Modified 05/05/2025

Course Description

This course addresses the high level of integration of graphics, programming, and dynamic data/information supplied by database management systems (DBMS). The coursework includes discussion and integration of graphics, still images, client and server side scripting languages and data supplied by a DBMS. A PHP/MySQL based electronic portfolio will be developed by each individual student during the course and posted to the Internet to provide potential employers with the student's resume, CIS Programming syllabi and samples of the student's computer programming work in HTML, CSS, JavaScript, PHP, MySQL, C# and Java. Prerequisite: CIS 130, CIS 196

Outcomes

Technical Skills

- WAMP Server
 - Utilize WAMP Server to run Apache and MySQL services.
- Basic PHP Skills
 - Utilize HTML/CSS/JS as appropriate to create a clean UI for end users
 - Utilize PHP include statements to import other files and create page templates
 - Utilize PHP control statements (if/else, try/catch, loops) to structure dynamic code
- PDO Objects and CRUD with MySQL Integration
 - Utilize a PDO Object to connect to a MySQL database with PHP
 - Allow web interface users to perform CRUD operations on the MySQL database
- MVC Pattern/PHP Functions
 - The MVC Pattern is used to create a modular application that is scalable.
 - Code is appropriately placed in Models, Controllers, and Views.
- PHP Sessions/Cookies

- Sessions are used to maintain and hold data for a user across a session.
- Cookies are used to hold persistent data across sessions.
- User Accounts/Logins
 - User Accounts are implemented to restrict functionality of the application to only authenticated users.
 - Actions/views are created to allow users to view and modify their information.
 - Login info and some other additional data is stored for users.
 - Passwords are properly stored as hashes, and logins check user passwords.

Professionalism/Communication

- Submission requirements for the assignment are met. This includes video length, file formats, etc.
- Version Control with GitHub
 - Tasks to be completed are created as issues and assigned within the GitHub repository, as well as resolved when completed.
 - Commits show incremental progress, and focus on adding features, fixing bugs, etc.
- Group Work
 - Students will collaborate and assign tasks within a group to complete a single application.

Critical Thinking

- Web Security
 - Applications are analyzed and best practices considered and implemented as appropriate.

Course Materials

Murach's PHP and MySQL

Author: Joel Murach, Ray Harris

Publisher: Murach

Edition: 3rd

ISBN: 978-1-943872-38-1

Deliverables

Homework

Weekly assignments will be used to track attendance. They will be a completion grade, and are intended as student self assessments. The instructor will not make corrections to the homework. It is intended that students study the material provided until they have a good understanding of it.

Unit Project

There are 3 units in this course, each with a Unit Project. Unit Projects will be the assessments for this course, where you will create an application to demonstrate skills, and then present the application to prove your skills. Presentations may be recorded if not attending class on presentation day.

In addition to the presentation, your instructor may require an interview on the Unit Projects to answer follow-up questions and gauge your understanding of the material.

Within Unit Projects, students will be assessed on competencies, which will be used to determine their Project Competencies - Approaching, Proficient, and Mastered scores.

✓ Evaluation Procedures and Grading

Criteria

Please Note: The intent of this grading system is for students to understand each topic in the course in order to pass. Completion of the homework and demonstrated mastery of the course topics is encouraged to raise your grade beyond a C, presuming that you are meeting proficiency in all of the competencies.

Type	Weight	Topic	Notes
Project Competencies - Approaching	40%		Student achieves at least a 2/4 on every project competency
Project Competencies - Proficient	30%		Student achieves at least a 3/4 on every project competency
Project Competencies - Mastery	10%		Each 4/4 score on project competencies, students will receive partial credit towards this score. Example: a student achieving 4/4 on half of the competencies will receive 5% of their final grade weight from this.
Participation/Homework	20%		Homework assignments weighted equally will give partial credit towards this score.

Breakdown

This breakdown is a model/example, not a hard rule for the final grading. Final grading will depend on the grade weighting as defined in the criteria.

Grade	Range	Notes
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Grade	Range	Notes
A	90%-100%	For an A, it is expected that students attend every class session , complete every competency with a score of 3/4 or higher, complete all homework, and completing some competencies with a score of 4/4.
B	80%-89%	For a B, it is expected that students achieve a 3/4 on every competency and also complete most homework assignments, and possibly receive a few 4/4 scores on competencies.
C	70%-79%	C is the minimum passing grade for this course. To receive a C in this course, a student must complete every competency on each assessment with a 3/4 or higher.
D	60% - 69%	Students participating completing all homework but not passing every competency with a 3/4 or higher may expect to receive a D in this course, if they complete every competency with a 2/4.



Course Outline

When	Topic	Notes
Unit 1 Weeks 1-3	Intro to PHP/MySQL	• WAMP Server • Basic PHP • Basic MySQL • PDO Objects
Unit 2 Weeks 4-7	Dynamic Web Applications	• The MVC Format • Web Forms • Arrays and Control Statements • Cookies and Sessions • Functions and Libraries
Unit 3 Weeks 8-10	Web Security	• Objects in PHP • Regular Expressions/Data Validation • Exception Handling • Reading/Writing Files • Secure Socket/Encryption • Login Systems